"Design and analysis of inverted AMOS sector antenna operating at 3 GHz"

Special Assignment Report

of Submitted in Partial Fulfilment of the

Requirements for completion of

Course on
2EC701 Microwave and Antenna Engineering

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November, 2024

CERTIFICATE

This is to certify that the Special Assignment Report entitled "Design and analysis of inverted AMOS sector antenna operating at 3 GHz" submitted by Mr Akshay Lakshmanan and Mr Aryan Makwana as per the academic records (21BEC059, 21BEC061) towards the partial fulfilment of for completion of Course on 2EC701 MICROWAVE AND ANTENNA ENGINNERING is the record of work carried out by him/her individually.

Date: 06/11/2024

Faculty Coordinator: Prof. Rutul Patel

Undertaking for Originality of the Work

I, Akshay Lakshmanan, 21BEC059, give undertaking that the Special Assignment Report entitled "Design and analysis of inverted AMOS sector antenna operating at 3 GHz" submitted by me, towards the partial fulfillment of for completion of Course on 2EC701 MICROWAVE AND ANTENNA ENGINNERING, is the original work carried out by me and I give assurance that no attempt of plagiarism has been made. I understand that in the event of any similarity found subsequently with any other published work or any report elsewhere; it will result in severe disciplinary action.

Akshay	**Aryan**
Signature of the Student 1	**Signature of the Student 2**

Date: 6 November 2024

Place: Ahmedabad

INDEX

1. INVERTED AMOS SECTOR ANTENNA SPECIFICATIONS

- **Category**: Aperture-type Antenna (Inverted Amos Sector)
- **Operating Frequency**: 3 GHz
- **Frequency Range**: The antenna can be simulated over a range of 1 GHz to 6 GHz to analyze its performance over a broad spectrum, focusing on behavior around the 3 GHz mark.
- **Directivity**: The antenna should be optimized to achieve a directivity value of approximately 5-6 dBi, providing a focused radiation pattern suitable for efficient signal transmission and reception in a specified direction.
- **Radiation Pattern**: The antenna is expected to exhibit a directional radiation pattern, which concentrates energy in a preferred direction. This characteristic is ideal for targeted communication applications where minimizing interference and maximizing signal strength in a specific area is crucial.

2. ANTENNA STRUCTURE AND DESIGN PARAMETERS

- **Ground Plane**: The ground plane is essential for the antenna's radiation pattern and performance. It acts as a reflector to direct the radiation forward.
- **Radiating Patch**: The primary radiating element is typically a sector-shaped patch. This patch has an angle resembling an inverted sector, often resembling a pie-slice shape. The dimensions of the patch define the resonant frequency and radiation characteristics.
- **Height (h)**: The distance between the patch and the ground plane affects the antenna's impedance and bandwidth. This height usually falls within a range that provides a good trade-off between bandwidth and gain.
- **Feed Point**: A coaxial feed or microstrip line is often used to excite the patch, which is placed at a calculated position to achieve the desired input impedance (often around 50 ohms).

3. SIMULATED ANTENNA

Using MATLAB, the antenna was simulated to observe its return loss, radiation pattern, and other key performance metrics. The antenna was modelled with the calculated dimensions, and a frequency sweep was conducted to examine its behaviour from 1 GHz to 10 GHz. The antenna was designed using inverted AMOS sector, where the sector ensures focused radiation and better impedance matching. The simulation setup included the following components:

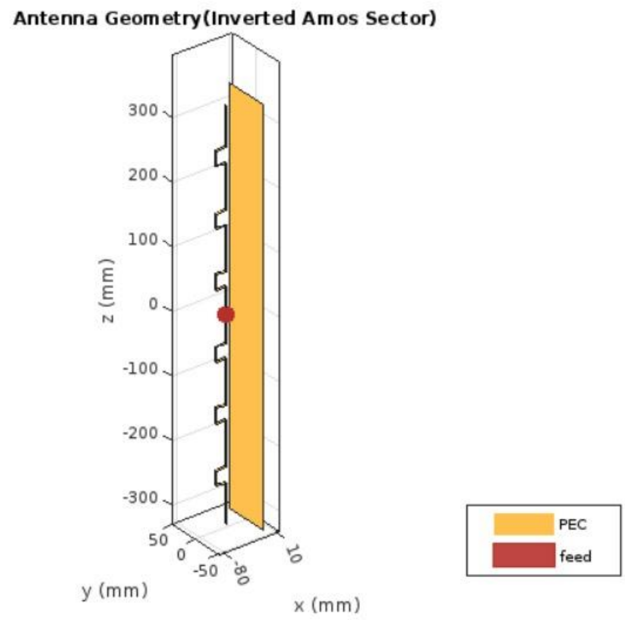

Fig.1 Inverted Amos Sector Antenna

4. Using the MATLAB App Designer for inverted AMOS Sector Antenna Analysis

This application was developed using MATLAB's App Designer, designed to simulate and analyse an inverted AMOS sector antenna's properties. The user can input various parameters to observe the antenna's performance at a specified frequency.

Steps to Use the Application

1. Launch the Application: Upon opening, the app displays input fields and buttons for user interaction.

2. Input Antenna Parameters:
   - Frequency: Enter the operating frequency in GHz.

3. Generate Patterns: Click the Load Pattern button to compute and display the following graphs based on the input parameters:
   - Return Loss: Plotted in the first axis, this graph represents the reflection coefficient across a frequency range.
   - Radiation Pattern: Shown in polar coordinates, illustrating the antenna's directional properties at the specified operating frequency.
   - Impedance v/s Frequency Chart: Displays the impedance matching of the antenna across the frequency range.

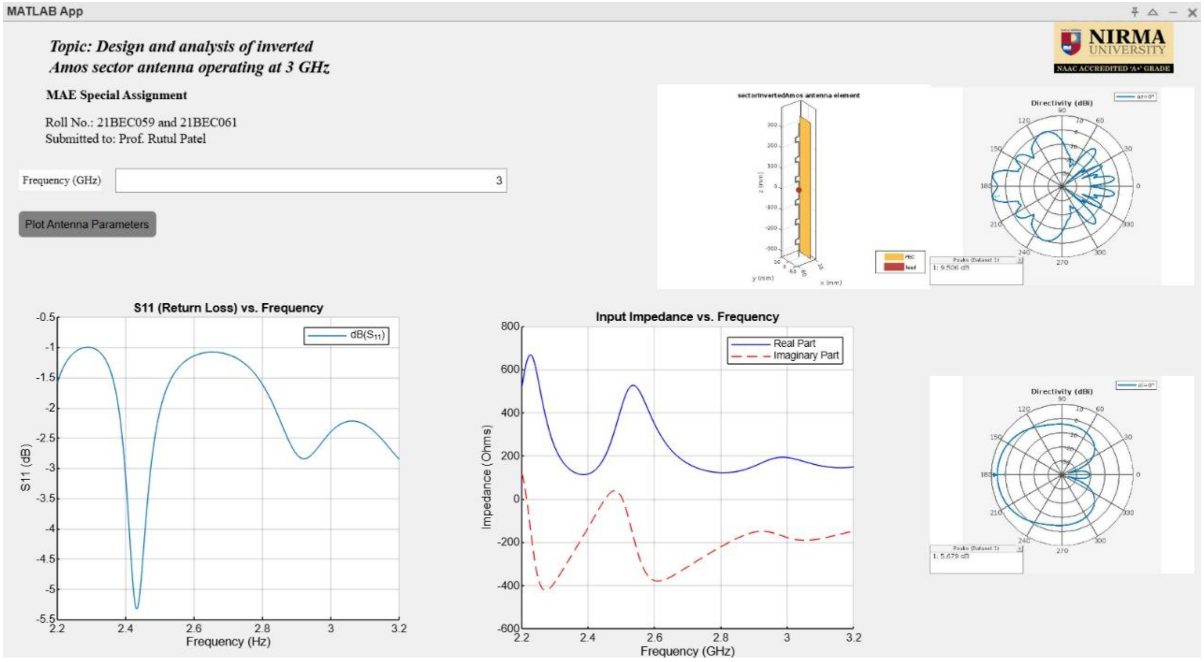

Fig.2 App Interface

5. RESULTS

The results from the simulation were analysed to ensure the antenna meets the design specifications.

Input Impedance vs. Frequency

The impedance plot exhibits variations in the real and imaginary components across 2.2 GHz to 3.2 GHz, with the real part reaching up to 600 ohms and the imaginary part crossing zero near 2.4 GHz and 3 GHz. These zero crossings correspond to resonance frequencies, where impedance becomes predominantly resistive, indicating efficient power transfer and minimal reactive losses at these points. This characteristic is essential for matching the antenna with a standard 50-ohm system, reducing mismatch losses around the operational frequency range.

Return Loss (S11) vs. Frequency

The S11 parameter, or return loss, shows a minimum of about -5 dB near 2.4 GHz, indicating moderate impedance matching at this frequency. Return loss values closer to -10 dB would indicate optimal matching, so further adjustments may be necessary for enhanced performance at 3 GHz. Improved matching will help reduce signal reflections, optimize energy radiation, and increase overall antenna efficiency at the desired operating frequency

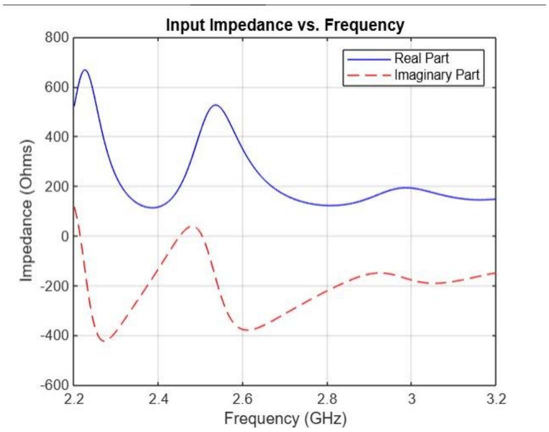

Fig.3 Input Impedance v/s Frequency Graph

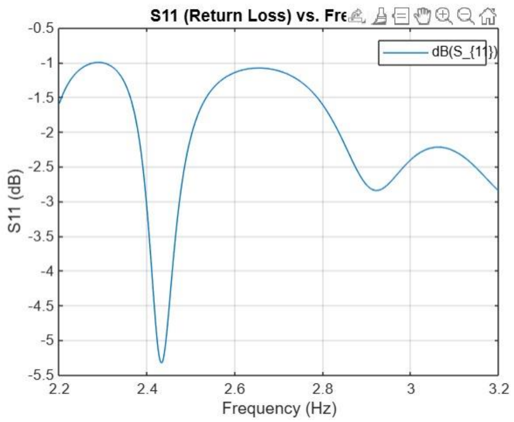

Fig.4 Return Loss v/s Frequency Graph

Directivity

The directivity pattern shows a peak gain of approximately 9.506 dBi, illustrating the antenna's ability to focus energy in a preferred direction. This high directivity implies efficient radiation in targeted areas, making it ideal for applications needing focused coverage, like sectorized communication systems. The pattern reveals strong forward lobes with reduced side and back lobes, which minimizes interference and enhances signal strength in the primary direction.

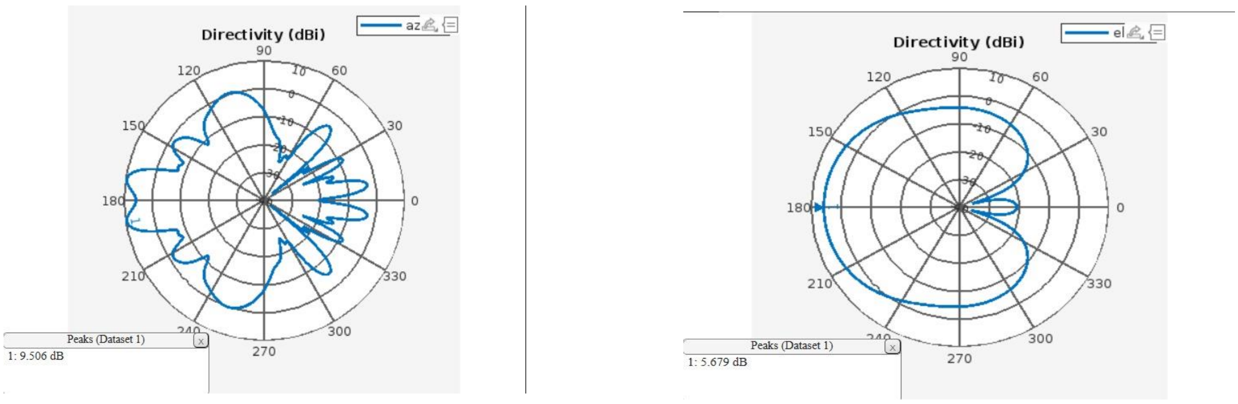

Fig.5 Directivity Graphs

6. CONCLUSIONS

The design of the inverted AMOS sector antenna focused on key elements like operating frequency, impedance matching, and radiation characteristics. After completion of this project a proper understanding of effects of antenna parameters on radiation pattern, return loss and impedance chart were observed. Using MATLAB App Designer for simulations and app designing, allowed us to fine-tune the antenna parameters with precision, leading to a design that performed well. The results showed minimal return loss at the desired frequency, a strong and focused radiation pattern, and good directivity, all of which confirmed the antenna's effectiveness.